MKT 4990 - Data Analytics Project

Marketing Data Analytics using Amazon E-commerce Data

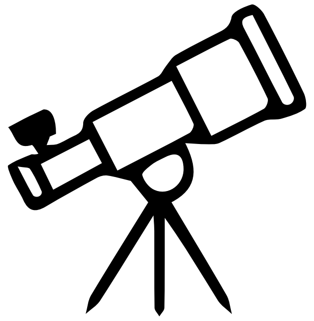
Implications to Retail Business Strategy

Turning raw data into Business insights

Presented by: Vaishnavi Perka - Data Science Graduate
Hariharan Jothimani - Data Science Graduate
Advisor: Dr. Junhong Min
Michigan Technological University · 2026

Why Does Data Matter in Business?

Businesses make thousands of decisions daily



What if we could predict customer behavior?

What if we knew who buys more, when, and why?

That's what this project explores.

Where Did Our Data Come From?

Amazon Transaction Dataset (1M+ records)

Purchase records, product categories, brands, quantities, and dates.

Customer Survey Dataset (5027 Users)

Demographics: Age, Gender, Income, Family Size, Account Usage, State.

Open e-Commerce 1.0 · Harvard Dataverse · doi:10.7910/DVN/YGLYDY

Key Fields Used: Age · Gender · Income · Purchase Quantity · Family Size · Account Usage · State

"We combined what people buy with who they are."

Reality of Data: It's Messy!

Missing Values

Fields left blank, sensors fail, users skip forms — gaps appear everywhere

Duplicate Records

System retries, merged imports, and human error create repeated entries

Inconsistent Formats

Dates as "01/03/24" vs "March 1, 2024", or "USA" vs "United States"

Outliers & Errors

A typo turns $50 into $50,000 — skewing every average and model

"In the real world, 80% of data work is cleaning — before any analysis begins."

Making Data Usable: Before vs After

METHOD	✗ BEFORE	✓ AFTER

Rename Columns	`Survey ResponseID`
	`ASIN/ISBN (Product Code)`
	`Shipping Address State`
		`User ID`
		`Product ID`
		`State`

Fill Missing Values	`Category 89,457 nulls`
	`Title 89,739 nulls`
	`State 87,811 nulls`
	`Product ID 973 nulls`
		`Replaced with “Unknown” Value`

Fix Date Data Type	`Order Date dtype: object`
	`(stored as plain text string)`
	`E.g. '2019-10-06'`
		`Year, Month, Day columns`
		`2019 10 06`

Flag Duplicates	`Row 33: R_01v... B07RTN4C77 $50.00`
	`Row 34: R_01v... B07RTN4C77 $50.00`
	`← 11,624 duplicate rows found`
		`Row 33: Flag (Yes)`
		`Row 34: Deleted`

Category Engineering

Transforming hundreds of raw product codes into 25 meaningful categories

BEFORE — Raw Categories

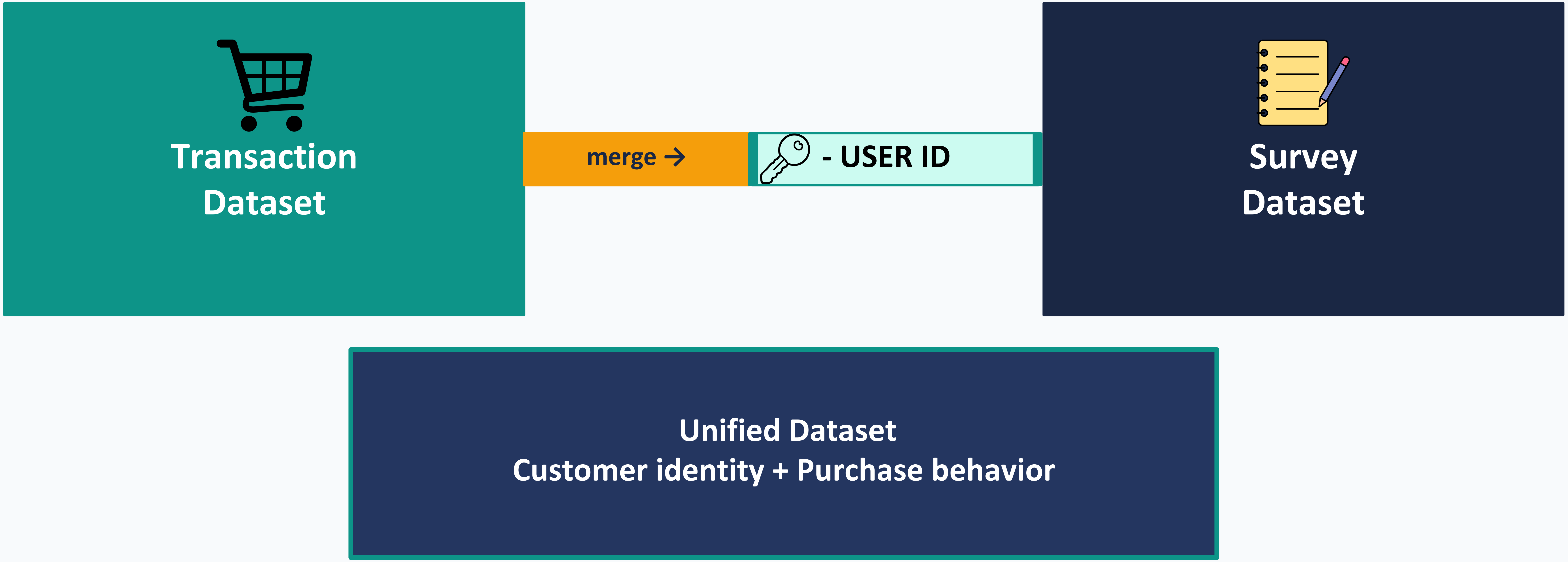
- LAPTOP_ACCESSORIES
- ACCELERATION_SENSOR
- MULTIPORT_HUB
- VDO_DEVICES
- ABIS_WIRELESS
- OPTICAL_DISC_DRIVE
- SATELLITE_OR_DSS
- SURVEILANCE_SYSTEMS
- GUITAR_STRINGS
- PACKING_CUBE
- *...400+ more codes*

→ *Keyword Matching*

AFTER — 25 Clean Categories

Electronics	Baby
Clothing	Jewelry
Beauty	Office Supplies
Food	Music & Media
Sports	Luggage & Travel
Health	Garden & Outdoor
Home-Bed-Bath	Industrial
Kitchen	Household Essentials
Toys	Software
Pet Supplies	Collectibles
Books	Misc
Automotive	
Hardware	

Combining Multiple Data Sources

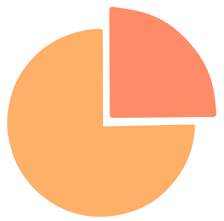


"Now we can connect who the customer is with what they buy."

Sample Dataset

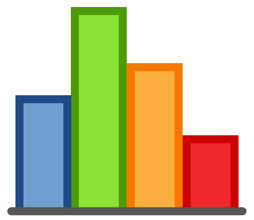
S.no	User_ ID	Prod uct	Cate gory	Title	Quan tity	Purc hase	State x	Dupli cate	Bran d	Year	Mont h	Day	Age	Hisp anic	Race	Educ ation	Inco me	Gend er	Sexu al Or	State y	No.of User	Famil y	Acc Usag	Cigar ettes	Marij uana	Alco hol	Diab etes	Whe el Ch	Life_ Chan	Sell_ Data	Sell_ Cons	Smal l Bus	Cens us U	Rese arch
1	R_01 vNla	B014 3RTB	Elect ronic	SanD isk	1	7.98	NJ	No	SanD isk	2018	Dec	4	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
2	R_01 vNla	B01 MA1	Elect ronic	Betr on	1	13.99	NJ	No	Betr on	2018	Dec	22	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
3	R_01 vNla	B078 JZTF	Unkn own	Unkn own	1	8.99	NJ	No	unkn own	2018	Dec	24	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
4	R_01 vNla	B06 XWF	Hom e-	Perfe cto	1	10.45	NJ	No	Perfe cto	2018	Dec	25	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
5	R_01 vNla	B00 837Z	Beau ty	Pror aso	1	10	NJ	No	Pror aso	2018	Dec	25	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
6	R_01 vNla	B01 GFB2	Elect ronic	Micr o	1	10.99	NJ	No	Micr o	2019	Feb	18	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
7	R_01 vNla	B00 NH13	Elect ronic	Amaz on	1	4.99	NJ	No	Amaz on	2019	Feb	18	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
8	R_01 vNla	B077 H6L7	Elect ronic	Fire HD 8	1	124.99	NJ	No	Fire HD	2019	Mar	15	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
9	R_01 vNla	B07L 84ZZ	Cloth ing	Men' s	1	12.99	NJ	No	Werf oru	2019	Apr	23	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
10	R_01 vNla	B06 XKN	Unkn own	Unkn own	1	24.69	NJ	No	unkn own	2019	Apr	23	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
11	R_01 vNla	B07C G71K	Elect ronic	UGR EEN	1	9.99	NJ	No	UGR EEN	2019	May	2	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
12	R_01 vNla	B079 GFF4	Elect ronic	Betr on	1	12.79	NJ	No	Betr on	2019	May	2	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
13	R_01 vNla	B01 MOO	Indu strial	NEW Norp	1	9.14	NJ	No	Norp ro	2019	May	11	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
14	R_01 vNla	B01 MA1	Elect ronic	Betr on	1	9.99	NJ	No	Betr on	2019	May	16	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
15	R_01 vNla	B073 JYC4	Elect ronic	SanD isk	1	19.99	NJ	No	SanD isk	2019	Jun	15	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
16	R_01 vNla	B00I 7DT4	Beau ty	Slipp ery	1	14.72	NJ	No	Slipp ery	2019	Jun	15	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
17	R_01 vNla	B01 MA1	Elect ronic	Betr on	1	9.99	NJ	No	Betr on	2019	Jun	25	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
18	R_01 vNla	B00 43T7	Elect ronic	Logit ech	1	26.77	NJ	No	Logit ech	2019	Jul	14	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes
19	R_01 vNla	B01 M2V	Elect ronic	Betr on	1	12.99	NJ	No	Betr on	2019	Aug	2	35 - 44	Yes	Black or	Bach elor's	$25, 000	Male	heter osex	NJ	1 (just	1 (just	Less than	No	No	No	No	No		Yes if I get	Yes if cons	No	No	Yes

Turning Data into Business Insights through Data Visualizations



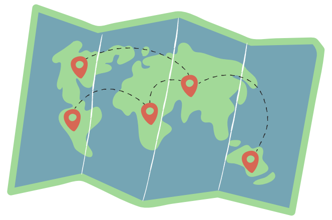
Pie Chart

Category distribution



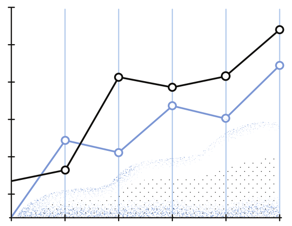
Bar Chart

Brand performance



Map

Geographic insights

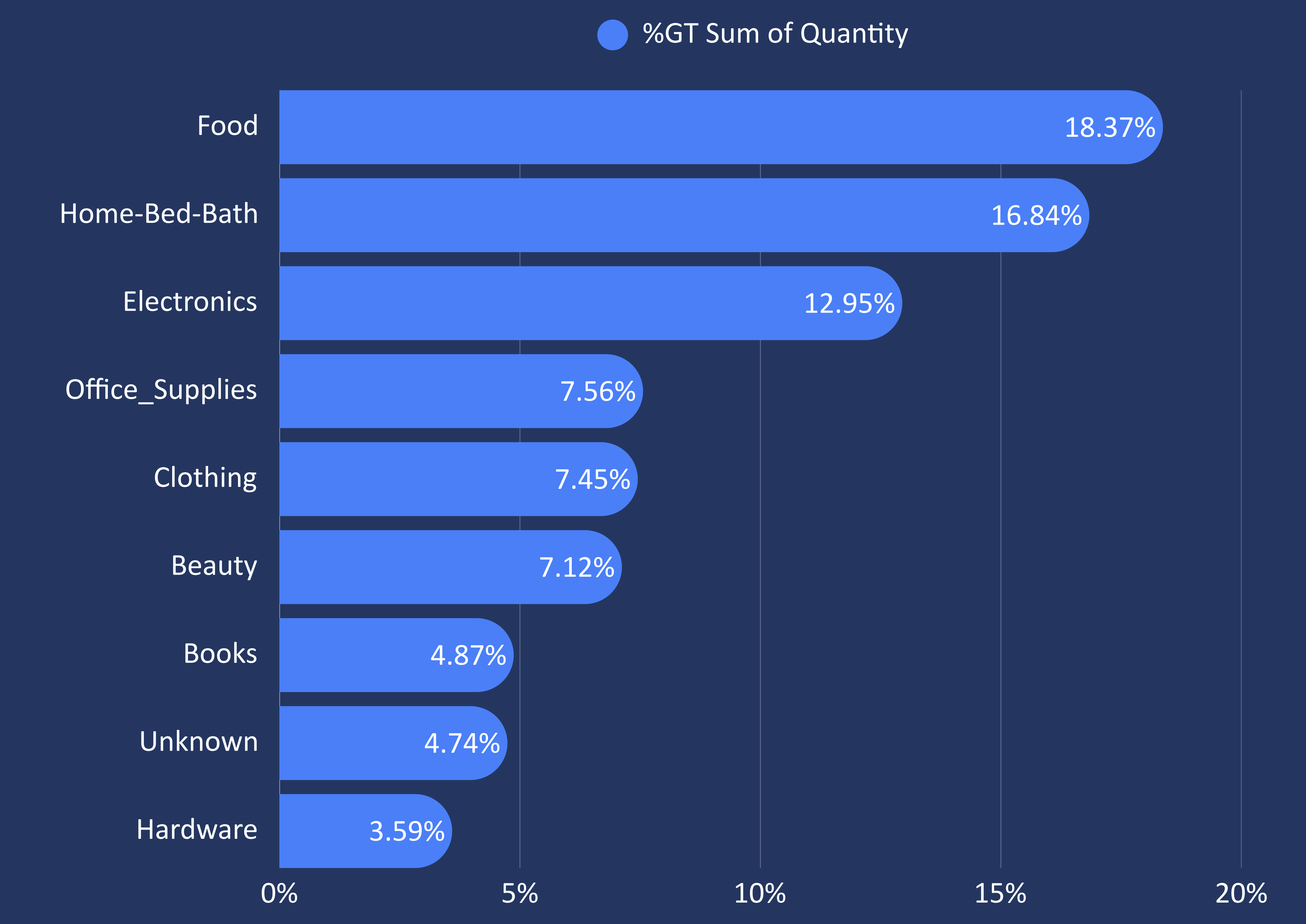


Line Chart

Trends over time

"Visuals help us see patterns instantly."

What Do Customers Buy the Most?



~18% - Food
~16% - Home-Bed-Bath

Top 2

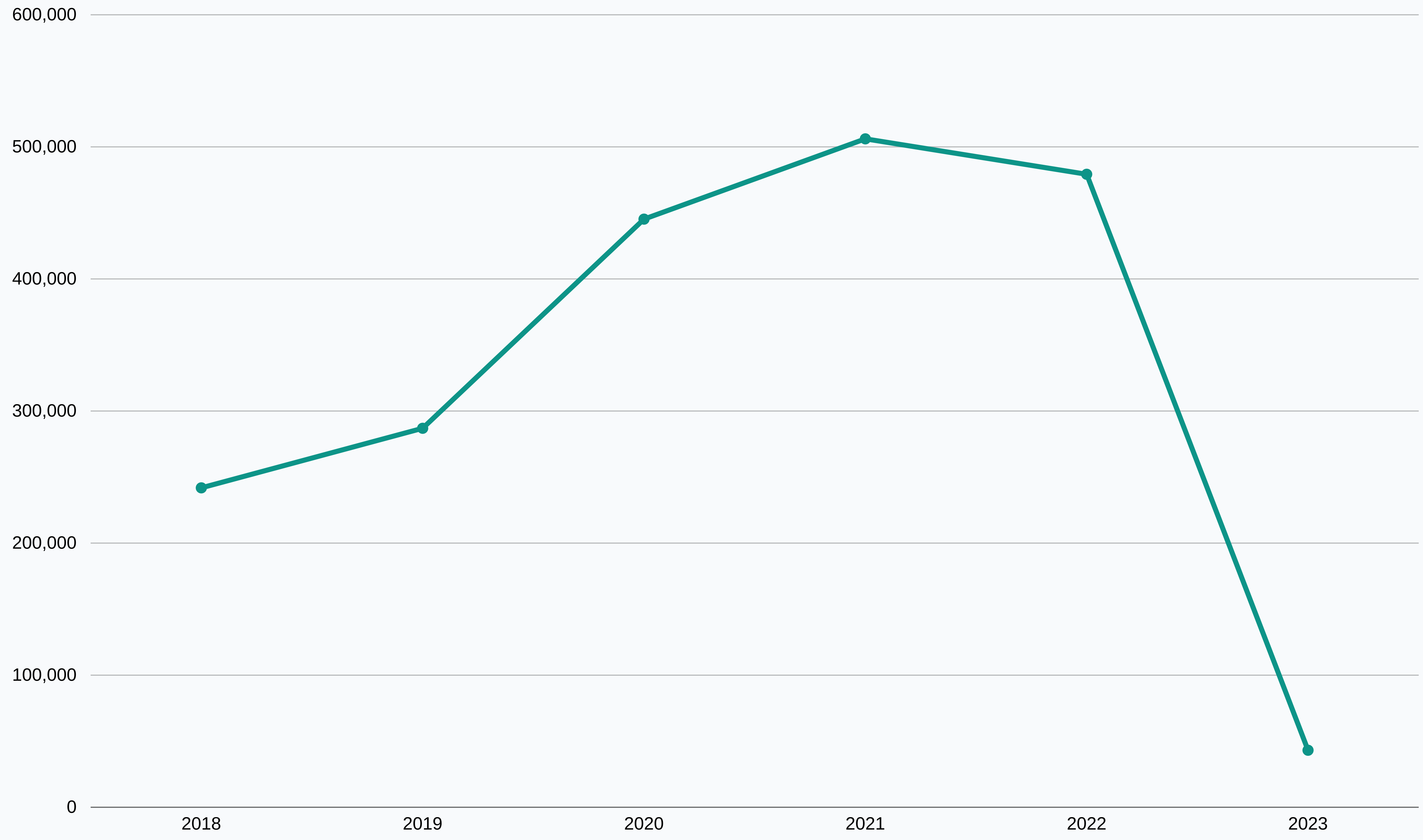
Categories are everyday essentials

↑

Demand is driven by necessity

"Essential products drive the highest demand."

How Purchases Changed Over Time



2018–2021
Steady growth phase

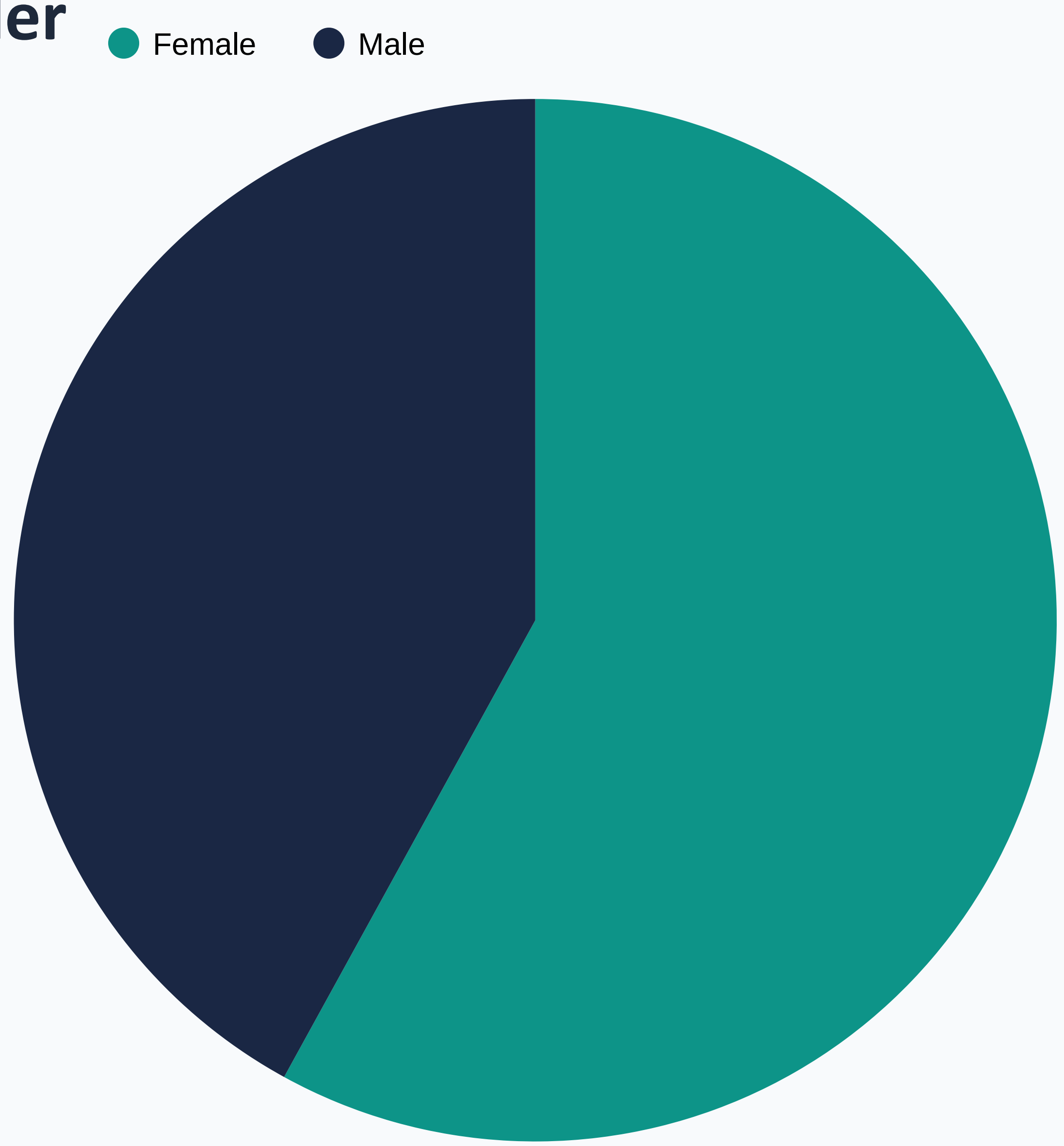
2020 Peak
Pandemic-driven demand

Post-2021 slight decline

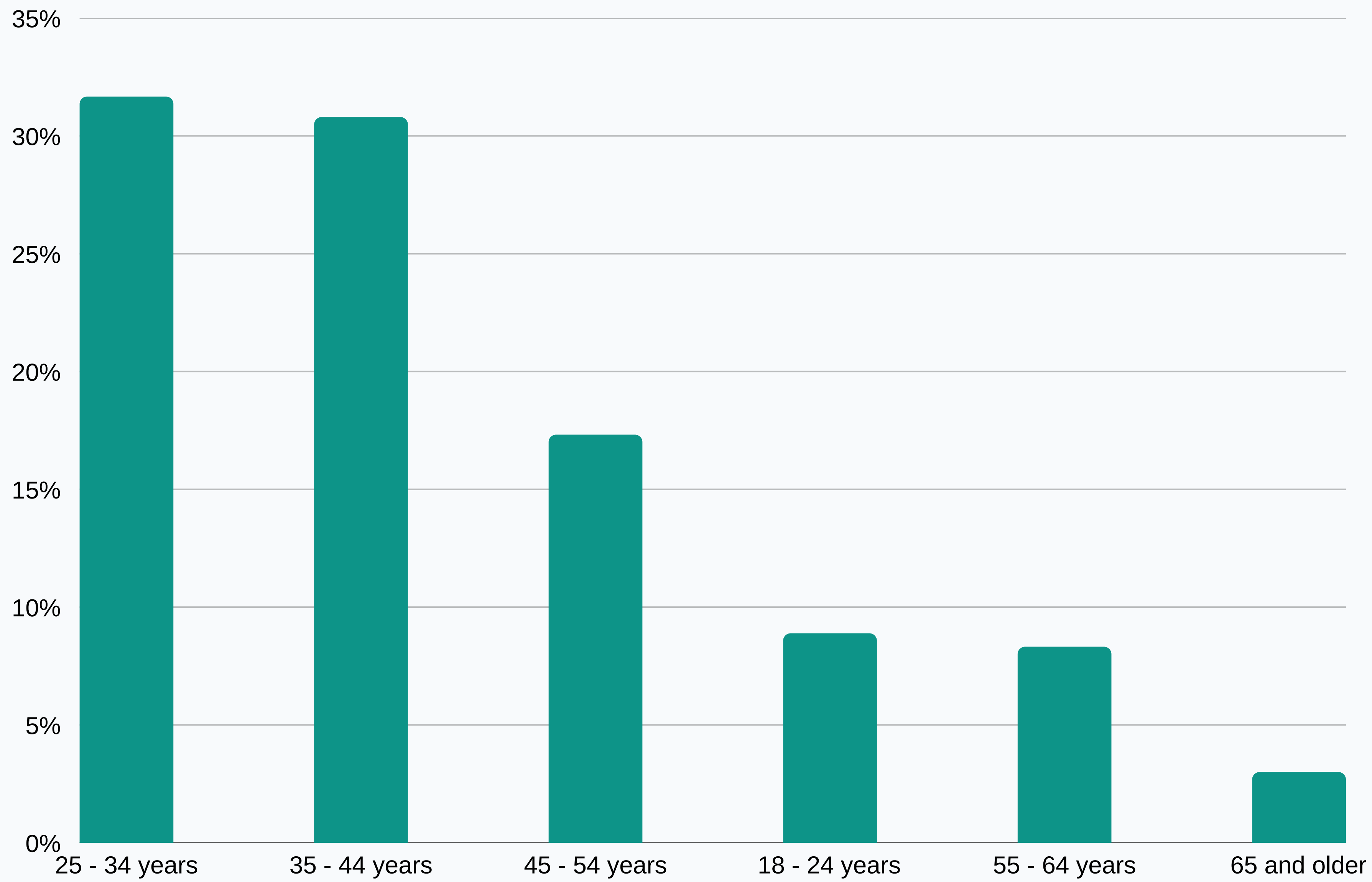
"Demand spiked significantly during peak years (likely pandemic effect)."

Who Buys More?

By Gender

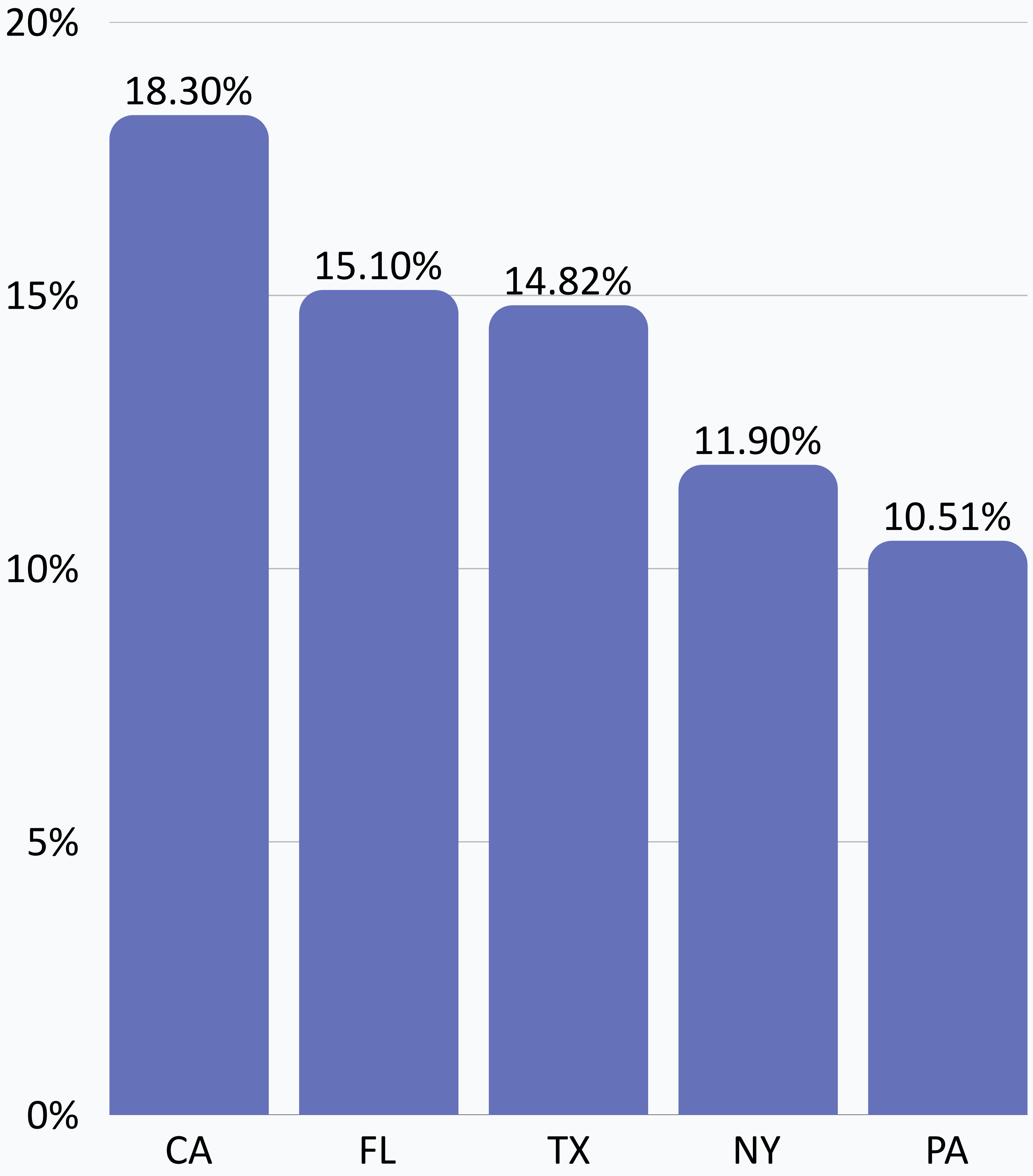


By Age Group

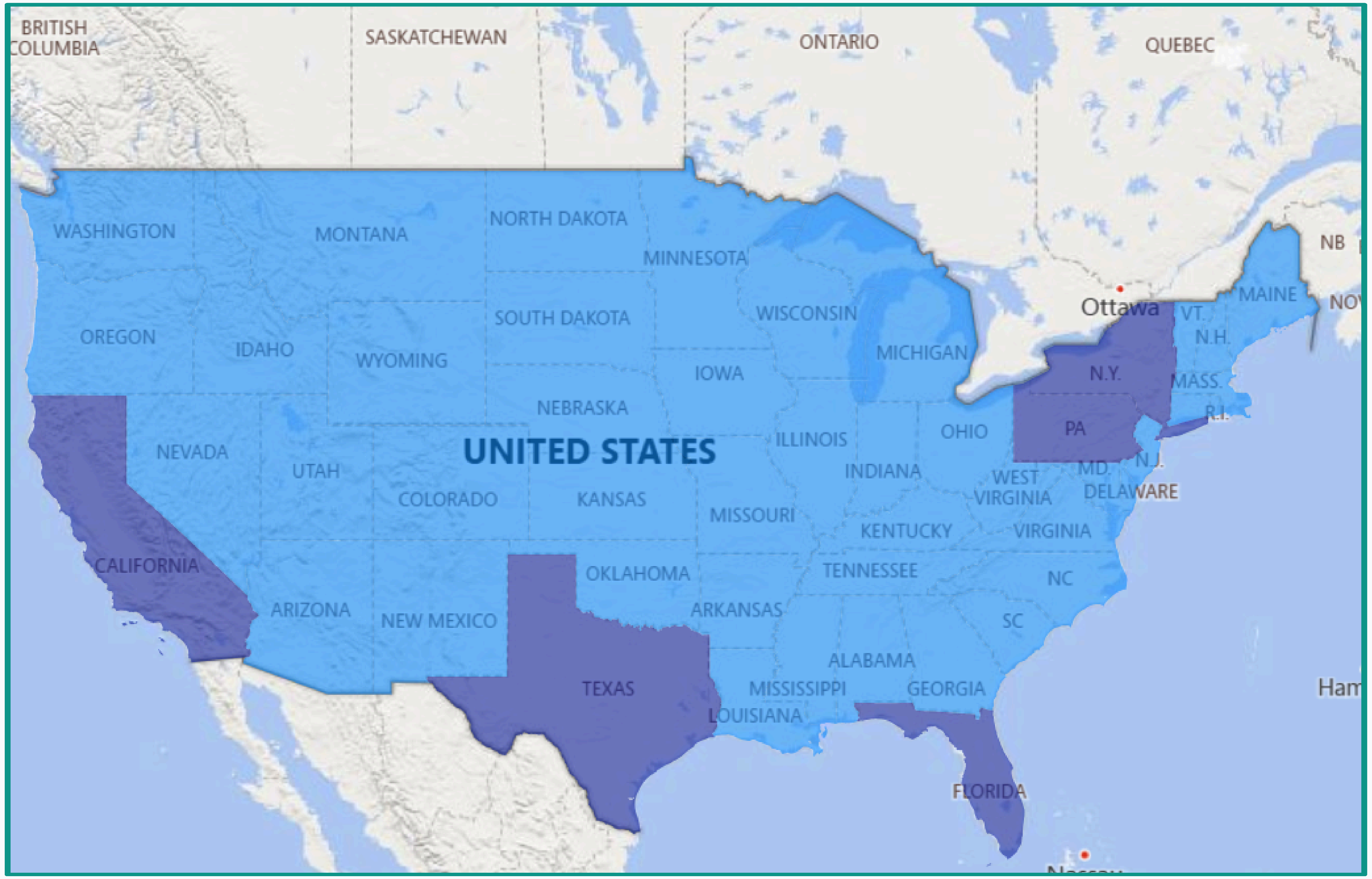


"Young adults (25–44) are the most active customers."

Where Are Customers Located?



Top States by number of users



Statistical Analysis: Overview

Three hypothesis tests were run on 1.8M+ merged transaction + survey records to test behavioral significance.

Test 1

Account Sharing vs Purchase Qty

One-Way ANOVA

Variables: No. of Users (1 / 2 / 3 / 4+)

All Significant ✓

Test 2

Family Size & Age vs Purchase Qty

Two-Way ANOVA

Variables: Family Size × Age Group

All Significant ✓

Test 3

App Usage & Income vs Purchase Qty

Two-Way ANOVA

Variables: Acc Usage × Income Level

All Significant ✓

Test 1: Account Sharing vs Purchase Quantity

H_0: No. of account users has NO significant effect on purchase quantity | Method: One-Way ANOVA

ANOVA Result

F-stat
5.477

p-value
0.0009

$p < 0.05$ ✓

Finding

Solo users drive the most purchases. Account sharing significantly affects buying behavior ($p < 0.001$).

Test 2: Family Size × Age Group vs Purchase Quantity

H_0: Family size & age group have NO significant effect on purchase quantity | Method: Two-Way ANOVA

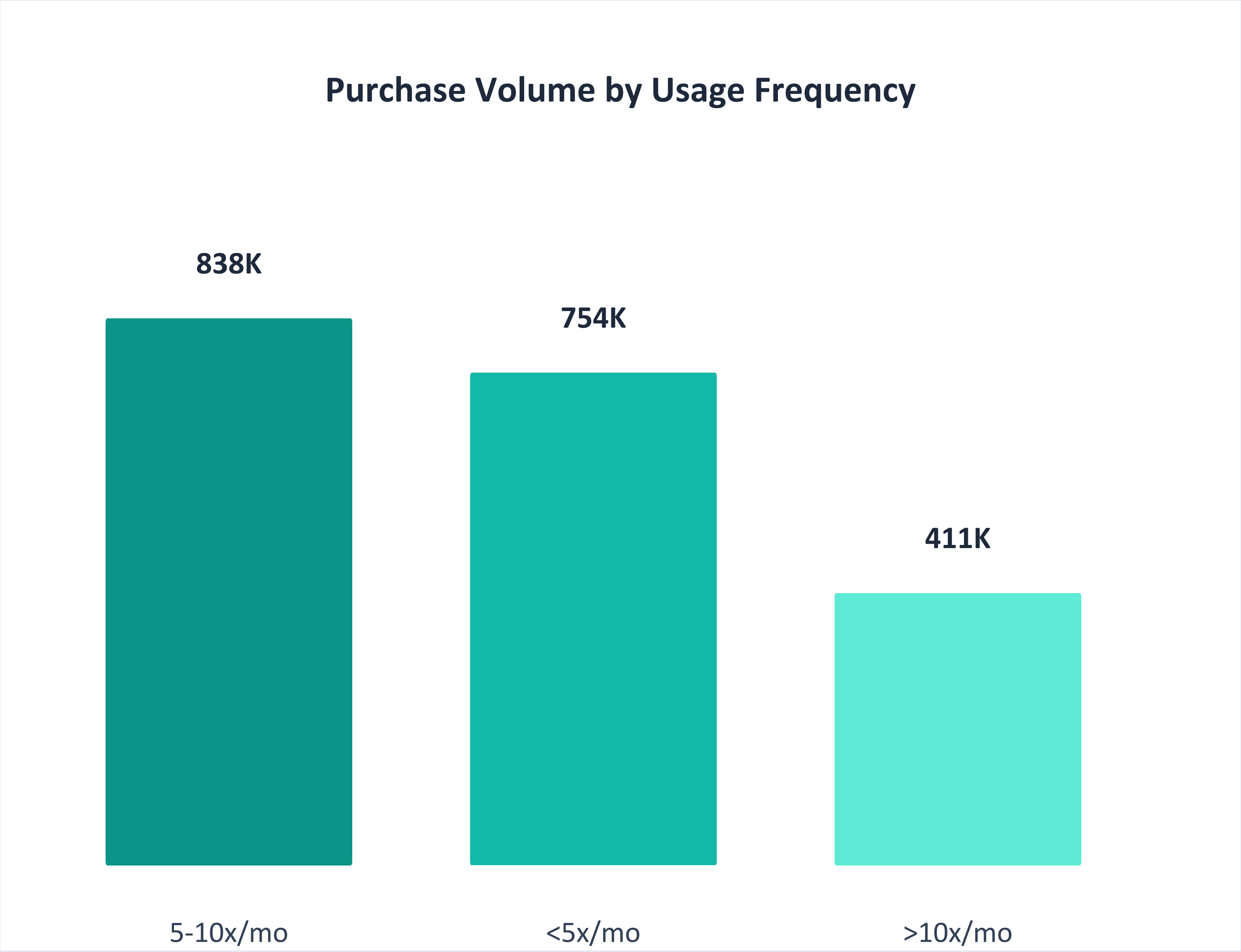
Factor	p-value	Significant?
Family Size	1.92e-11	Yes ✓
Age Group	1.12e-144	Yes ✓
Interaction (Family × Age)	2.28e-30	Yes ✓

Key Findings

- Family size alone significantly affects purchase quantity ($F=17.6$, $p<0.001$)
- Age group has the strongest independent effect ($F=136$, $p \approx 10^{-144}$)
- Interaction effect is also significant — age moderates how family size influences buying
- 2-person households make the highest total purchases

Test 3: App Usage × Income vs Purchase Quantity

H_0: App usage frequency & income level have NO significant effect on purchase quantity | Method: Two-Way ANOVA



Factor	p-value	Significant?
Acc Usage	2.58e-04	Yes ✓
Income Level	1.70e-37	Yes ✓
Interaction (Usage × Income)	4.64e-28	Yes ✓

Key Findings

- Usage frequency has a modest but significant effect on quantity ($F=8.26$, $p<0.001$)
- Income level drives the strongest differences in buying volume ($F=31.0$)
- Interaction is significant — income moderates how usage affects purchasing
- Bonus: Pearson r(usage, qty) ≈ 0.000 → usage alone is nearly uncorrelated

Business Impact to Amazon

Target Key Customers

- Customers aged 25–44 drive the majority of purchases
- Invest in personalized recommendations & targeted ads for this segment
- Female buyers slightly outnumber males — tailor messaging accordingly

Prioritize Top Categories

- Food (18%) and Home-Bed-Bath (16%) dominate demand — stock & promote heavily
- Electronics (13%) is third — highlight deals and bundles
- Seasonal promotions in top categories can maximize revenue

Data-Driven Brand Management

- "Unknown" brands in 4.74% of records lost attribution insights
- Enforce complete brand tagging in seller onboarding to enable brand analytics
- Use clean brand data to identify top performers and surface private-label gaps

Avoid Wrong Assumptions

- Shared accounts (2+ users) buy LESS than solo accounts — rethink family plan strategy
- Heavy app usage alone doesn't drive purchases; income moderates the effect
- Post-2021 demand decline signals need for re-engagement campaigns

Recommendations for Houghton Retailers

Insights from Amazon data applied to your local Houghton market

Know Your Customer (MTU Students + Locals)

- 25–44 age group dominates Amazon purchases — MTU students & young professionals are your prime segment
- Offer student discounts, loyalty cards, and bundle deals to capture this high-frequency group
- Omni-channel strategy: bridge online awareness with in-store experience

Stock What Amazon Can't Beat

- Offer specialty & locally-sourced products Amazon doesn't carry (e.g., UP/Michigan-themed goods)
- Bath & Body — compete with niche/natural brands; capitalize on Home-Bed-Bath demand trend
- Running & outdoor gear downtown: align with MTU's active campus lifestyle

Go Digital — Benchmark Against Amazon

- Digitize receipts and sales data to track your own purchase trends
- Use Keepa API or Amazon bestseller data to benchmark pricing and category demand
- Build social media presence: Awareness → Engagement → Conversion funnel

Leverage Local Loyalty

- Houghton is a small, tight-knit community — locals are highly loyal to trusted in-town stores
- Don't assume small market = low opportunity; loyalty translates to repeat revenue
- Invest in community events, partnerships with MTU clubs, and local sponsorships

What Can Be Done Next?

ML Prediction Models

Predict customer purchases before they happen using machine learning

Personalized Recommendations

Suggest products tailored to each customer's profile

Real-Time Dashboards

Live data monitoring for instant business decisions

Customer Segmentation

Group customers into meaningful clusters for targeted campaigns

"We can move from analysis → prediction → personalization."

Final Takeaways

- Data tells hidden stories that intuition alone cannot reveal
- Cleaning and preprocessing data is critical before any analysis
- Visualizations simplify complex patterns for faster understanding
- Statistics validate insights — ensuring patterns are real, not random
- Data-driven decisions are smarter, faster, and more reliable

Thank You!

Questions & Discussion